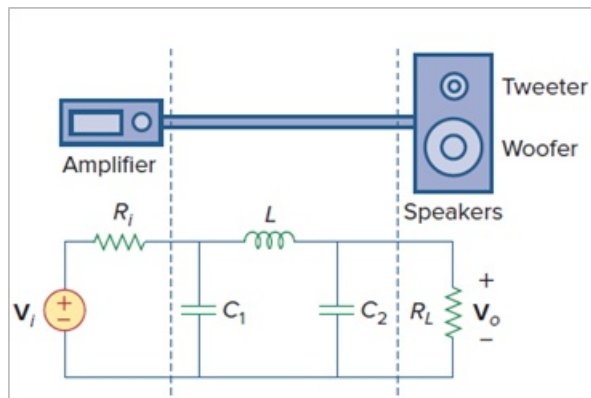


Problem

The crossover circuit in Fig. 14.108 is a low-pass filter that is connected to a woofer. Find the transfer function $\mathbf{H}(\omega) = \mathbf{V}_o(\omega)/\mathbf{V}_i(\omega)$.

Figure 14.108:



Step-by-step solution

Step 1 of 7

Refer to section 14.12, and Figure 14.108 in the textbook.

Consider the following circuit diagram of a low pass filter as shown in Figure 1.

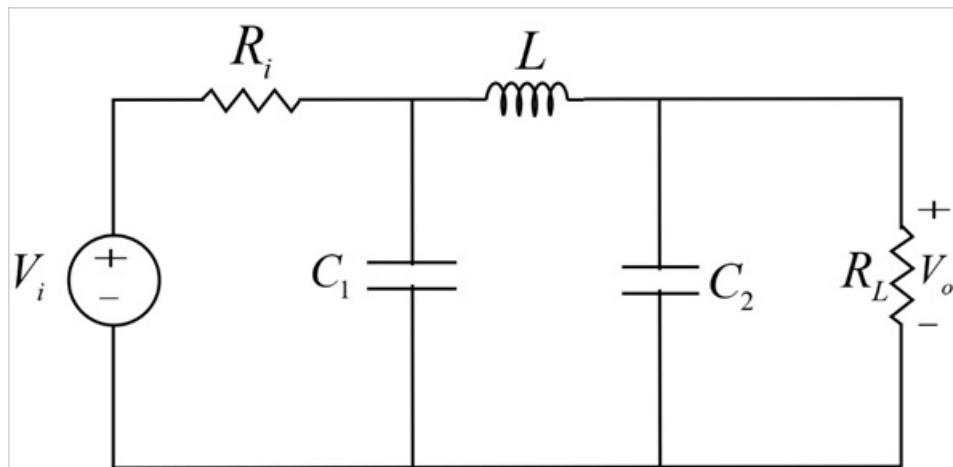


Figure 1

Step 2 of 7

Now consider the modified circuit diagram in frequency domain as shown in Figure 2.

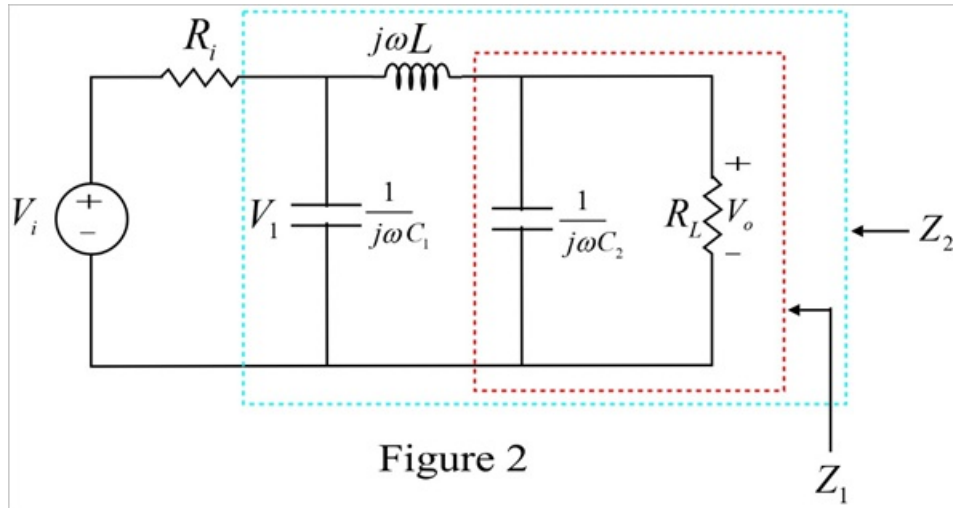


Figure 2

Step 3 of 7

Let V_1 be the voltage across capacitor as shown in Figure 2.

Determine the impedance Z_2 .

Step 4 of 7

Determine the impedance Z_2 .

Substitute V_1 .

$$\begin{aligned}
 Z_2 &= \frac{\left(\frac{1}{j\omega C_1} \right) \left(j\omega L + \frac{R_L}{1 + j\omega R_L C_2} \right)}{\left(\frac{1}{j\omega C_1} \right) + \left(j\omega L + \frac{R_L}{1 + j\omega R_L C_2} \right)} \\
 &= \frac{\left(\frac{1}{j\omega C_1} \right) \left(\frac{j\omega L(1 + j\omega R_L C_2) + R_L}{1 + j\omega R_L C_2} \right)}{\left(\frac{1}{j\omega C_1} \right) + \left(\frac{j\omega L(1 + j\omega R_L C_2) + R_L}{1 + j\omega R_L C_2} \right)} \\
 &= \frac{\left(\frac{j\omega L(1 + j\omega R_L C_2) + R_L}{(j\omega C_1)(1 + j\omega R_L C_2)} \right)}{\left(\frac{(1 + j\omega R_L C_2) + (j\omega C_1)(j\omega L(1 + j\omega R_L C_2) + R_L)}{(j\omega C_1)(1 + j\omega R_L C_2)} \right)} \\
 &= \frac{j\omega L(1 + j\omega R_L C_2) + R_L}{(1 + j\omega R_L C_2) + (j\omega C_1)(j\omega L(1 + j\omega R_L C_2) + R_L)} \\
 Z_2 &= \frac{j\omega L - \omega^2 R_L L C_2 + R_L}{1 + j\omega R_L C_2 + (j\omega C_1)(j\omega L - \omega^2 R_L L C_2 + R_L)} \\
 &= \frac{j\omega L - \omega^2 R_L L C_2 + R_L}{1 + j\omega R_L C_2 - \omega^2 L C_1 - j\omega^3 R_L L C_1 C_2 + j\omega R_L C_1}
 \end{aligned}$$

Step 5 of 7

Consider the expression for the voltage $V_1(\omega)$ using voltage division rule.

$$V_1(\omega) = \left(\frac{Z_2}{Z_2 + R_i} \right) V_i(\omega) \dots\dots (1)$$

Step 6 of 7

Determine the voltage $V_o(\omega)$ using voltage division rule.

$$V_o(\omega) = \left(\frac{Z_1}{Z_1 + j\omega L} \right) V_1(\omega)$$

Substitute $\left(\frac{Z_2}{Z_2 + R_i} \right) V_i(\omega)$ for $V_1(\omega)$ using equation (1).

$$V_o(\omega) = \left(\frac{Z_1}{Z_1 + j\omega L} \right) \cdot \left(\frac{Z_2}{Z_2 + R_i} \right) V_i(\omega)$$

$$\frac{V_o(\omega)}{V_i(\omega)} = \left(\frac{Z_1}{Z_1 + j\omega L} \right) \cdot \left(\frac{Z_2}{Z_2 + R_i} \right)$$

Step 7 of 7

Determine the transfer function $H(\omega)$.

$$H(\omega) = \frac{V_o(\omega)}{V_i(\omega)}$$

$$H(\omega) = \left(\frac{Z_1}{Z_1 + j\omega L} \right) \cdot \left(\frac{Z_2}{Z_2 + R_i} \right)$$

Substitute , and $Z_2 = \left(\frac{j\omega L - \omega^2 R_L L C_2 + R_L}{1 + j\omega R_L C_2 + -\omega^2 L C_1 - j\omega^3 R_L L C_1 C_2 + j\omega R_L C_1} \right)$.

Therefore, the required value of the transfer function is,

.